

Disk Scheduling Algorithms: FCFS, SSTF and SCAN Performance Analysis

A Comprehensive Theoretical, Mathematical, and Graphical Simulation Study with Step-by-Step Trajectory Verification

Assigned Student:	Akshat Joshi
Course Curriculum:	Operating Systems (Unit VI: Storage Management)
Project Category:	Special Direct Reading Project (25 Points Allocation)
Primary Algorithms:	FCFS (First-Come, First-Served), SSTF (Shortest Seek Time First), SCAN (Elevator)
Extended Algorithms:	C-SCAN (Circular SCAN), LOOK, C-LOOK
Theoretical Benchmark:	Silberschatz et al. (10th Ed.) Benchmark: [98, 183, 37, 122, 14, 124, 65, 67] at Head 53
Core Deliverables:	Interactive GUI, Mechanical Platter Simulation, Trajectory Graphs, Calculations, Comparison Matrix
Academic Session:	2026–2027 Academic Year

Project Abstract: Secondary storage devices, specifically electromechanical magnetic hard disks, represent the primary I/O bottleneck in modern computer architectures. Because physical head displacement (seek latency) dominates disk access time by several orders of magnitude over electronic memory transfers, the disk scheduling subsystem within the Operating System kernel plays a critical role in system throughput, latency, and fairness.

This project presents a high-fidelity interactive simulation software and comparative performance analysis implementing the required syllabus algorithms (FCFS, SSTF, SCAN) alongside extended variants (C-SCAN, LOOK, C-LOOK). Every algorithm is mathematically modeled, visually simulated with real-time platter and actuator kinematics, evaluated against standard textbook benchmark problems, and benchmarked across Total Head Movement (THM), Average Seek Length (ASL), seek latency, starvation susceptibility, and mechanical arm stress.

1. Executive Summary & Expected Deliverables

Operating systems must arbitrate access to secondary storage when multiple processes generate simultaneous read and write requests. In accordance with the official Operating Systems project guidelines, this work delivers:

- **Rigorous Theoretical Exposition:** Mathematical formulation of disk latency parameters and scheduling heuristics.
- **High-Fidelity Graphical Simulation (GUI):** Live interactive magnetic platter animation with concentric cylinder tracks, pivoting actuator arm, linear cylinder ruler (0 to Max), and step-by-step playback controls.
- **Step-by-Step Calculation Engine:** Complete mathematical breakdowns showing absolute seek distances ($|C_i - C_{i-1}|$), cumulative Total Head Movement (THM), and operational rationale.
- **Multi-Dimensional Graphical Visualizations:** Silberschatz 2D Trajectory line plots (Cylinder vs Step Order) and comparative bar charts.
- **Comparative Performance Matrix:** Comprehensive evaluation table comparing THM, Average Seek Length (ASL), seek time, efficiency vs baseline FCFS, starvation risk, and algorithmic time complexity.

2. Hard Disk Physical Mechanics & Mathematical Formulations

A Hard Disk Drive (HDD) consists of spinning magnetic platters mounted on a common spindle rotating at constant speed (e.g., 5400, 7200, or 15000 RPM). Platters are divided into concentric circular **tracks**, and tracks aligned vertically across all platter surfaces form a **cylinder**. A mechanical actuator arm moves the read/write head assembly radially across cylinders.

Components of Disk Access Time:

The total latency incurred when servicing a disk read/write operation is expressed as:

$$T_{\text{access}} = T_{\text{seek}} + T_{\text{rotational}} + T_{\text{transfer}} + T_{\text{controller}}$$

- **Seek Time (T_{seek}):** The time required for the actuator arm to mechanically position the head over the target cylinder track. *This is the primary mechanical delay optimized by scheduling algorithms.*
- **Rotational Latency ($T_{\text{rotational}}$):** The delay waiting for the desired sector to rotate under the head (average = $1 / (2 \times \text{RPM})$).
- **Transfer Time (T_{transfer}):** The time needed to magnetically stream the data bits between the platter surface and disk controller buffer.

Mathematical Evaluation Metrics:

Evaluation Metric	Mathematical Formula	Operational Significance
Total Head Movement (THM)	$THM = \sum_{i=1..N} C_i - C_{i-1} $	Total cylinder tracks traversed by the read/write head from initial head C_0 across all N requests.
Average Seek Length (ASL)	$ASL = THM / N$	Mean track displacement per I/O operation. Directly indicates expected mechanical travel per request.
Estimated Seek Time (T_{seek})	$T_{\text{seek}} = THM \times t_{\text{track}}$	Total mechanical seek latency in ms, using track-to-track latency factor t_{track} (default: 5.0 ms/track).
Percentage Efficiency Gain (η)	$\eta = ((THM_{\text{FCFS}} - THM_{\text{alg}}) / THM_{\text{FCFS}}) \times 100\%$	Quantitative reduction in head travel achieved relative to the unoptimized baseline FCFS.

3. Theoretical Analysis of Disk Scheduling Algorithms

3.1 First-Come, First-Served (FCFS)

Mechanism: Requests are serviced strictly in the chronological order of their arrival into the I/O queue. The controller makes no attempt to reorder requests or optimize head trajectory.

- **Strengths:** Complete fairness across all processes; mathematically immune to starvation; $O(1)$ queue insertion.
- **Weaknesses:** Prone to severe wild swings across the platters, resulting in extremely high Total Head Movement and high mechanical stress.

3.2 Shortest Seek Time First (SSTF)

Mechanism: A greedy heuristic that selects the pending request closest to the current head position: $Next = \underset{r \in Q}{argmin} |C_{current} - r|$. Direction is preserved during ties.

- **Strengths:** Dramatically reduces Total Head Movement compared to FCFS, maximizing immediate I/O throughput.
- **Weaknesses:** **Starvation:** Continuous arrivals of localized requests near the head will indefinitely postpone requests residing on distant tracks.

3.3 SCAN (The Elevator Algorithm)

Mechanism: The head begins at its current position and moves in a specified direction (e.g., towards higher cylinders), servicing all requests encountered along its path. Upon reaching the physical disk boundary (track 199 or 0), the arm reverses direction and sweeps back.

- **Strengths:** Eliminates starvation entirely; delivers bounded worst-case waiting times; smooth monotonic arm motion.
- **Weaknesses:** Redundant travel to the physical boundary (track 199) even when no requests exist there; slight unfairness for requests just behind the reversing arm.

3.4 Extended Algorithms: C-SCAN, LOOK, and C-LOOK

- **C-SCAN (Circular SCAN):** Provides uniform waiting times by servicing requests in one direction only. Upon reaching the boundary, the arm immediately returns to the starting boundary *without servicing requests on the return trip*.
- **LOOK:** An optimization over SCAN where the arm reverses immediately upon servicing the final pending request in the current direction, completely avoiding unnecessary travel to the physical disk limits.
- **C-LOOK:** Combines C-SCAN's circular fairness with LOOK's boundary optimization, snapping back from the highest requested cylinder directly to the lowest requested cylinder.

Summary of Algorithmic Trade-Offs:

Algorithm	Heuristic Basis	Boundary Behavior	Starvation Risk	Complexity
FCFS	FIFO arrival order	No boundary awareness	Zero (Fair)	$O(N)$
SSTF	Min $ C_{curr} - r $ greedy	Opportunistic seek	High	$O(N^2)$
SCAN	Bidirectional sweep	Travels to physical limit (0 / Max)	None	$O(N \log N)$
C-SCAN	Unidirectional circular sweep	Resets from limit to limit	None (Uniform)	$O(N \log N)$
LOOK	Bidirectional bounded sweep	Reverses at extreme request	None	$O(N \log N)$
C-LOOK	Unidirectional bounded sweep	Resets to lowest request	None (Uniform)	$O(N \log N)$

4. Benchmark Dataset & Mathematical Step Calculations

Silberschatz et al. (10th Edition) Standard Problem Parameters:

- Total Cylinders: **0 to 199** (200 tracks) | Initial Head Position: **53** | Arm Direction: **Right (towards 199)**
- I/O Request Queue Sequence: **[98, 183, 37, 122, 14, 124, 65, 67]**

4.1 FCFS Step Calculations (THM = 640, ASL = 80.00)

Step #	From	To	Formula	Seek (Δ)	Cum. THM	Operational Rationale
1	53	98	98 - 53	45	45	1st request in FIFO queue
2	98	183	183 - 98	85	130	Large outward swing to track 183
3	183	37	37 - 183	146	276	Massive inward reversal across platters
4	37	122	122 - 37	85	361	Outward swing back towards outer tracks
5	122	14	14 - 122	108	469	Severe inward swing to track 14
6	14	124	124 - 14	110	579	Severe outward reversal to track 124
7	124	65	65 - 124	59	638	Inward seek towards middle tracks
8	65	67	67 - 65	2	640	Final nearby sequential request

4.2 SSTF Step Calculations (THM = 236, ASL = 29.50, 63.1% Gain)

Step #	From	To	Formula	Seek (Δ)	Cum. THM	Operational Rationale
1	53	65	65 - 53	12	12	Nearest cylinder to initial head 53
2	65	67	67 - 65	2	14	Nearest cylinder to 65 (67-65 =2 vs 37-65 =28)
3	67	37	37 - 67	30	44	Nearest remaining cylinder (37-67 =30 vs 98-67 =31)
4	37	14	14 - 37	23	67	Nearest remaining cylinder to 37
5	14	98	98 - 14	84	151	Left side exhausted; seek to nearest right
6	98	122	122 - 98	24	175	Nearest cylinder to 98
7	122	124	124 - 122	2	177	Adjacent cylinder seek
8	124	183	183 - 124	59	236	Final remaining outermost cylinder

4.3 SCAN Step Calculations (THM = 331, ASL = 41.38, 48.3% Gain)

Step #	From	To	Formula	Seek (Δ)	Cum. THM	Operational Rationale
1	53	65	65 - 53	12	12	Sweeping Right: first request above 53
2	65	67	67 - 65	2	14	Sweeping Right: next request along path
3	67	98	98 - 67	31	45	Sweeping Right: servicing cylinder 98
4	98	122	122 - 98	24	69	Sweeping Right: servicing cylinder 122
5	122	124	124 - 122	2	71	Sweeping Right: servicing cylinder 124
6	124	183	183 - 124	59	130	Sweeping Right: highest request serviced
7	183	199	199 - 183	16	146	Reaches physical boundary limit (199); reverses
8	199	37	37 - 199	162	308	Sweeping Left: first pending request encountered
9	37	14	14 - 37	23	331	Sweeping Left: final remaining inner request

5. Interactive Simulator GUI & Physical Disk Simulation

To bring disk scheduling concepts alive, a high-fidelity simulator was built using modern web standards. The application provides real-time animated physical disk mechanics, dynamic linear ruler tracking, and custom sequence builders.

Figure 1: Disk Parameters & Request Sequence Configuration Panel



Figure 2: Physical Disk Platter Assembly & Linear Cylinder Track Ruler

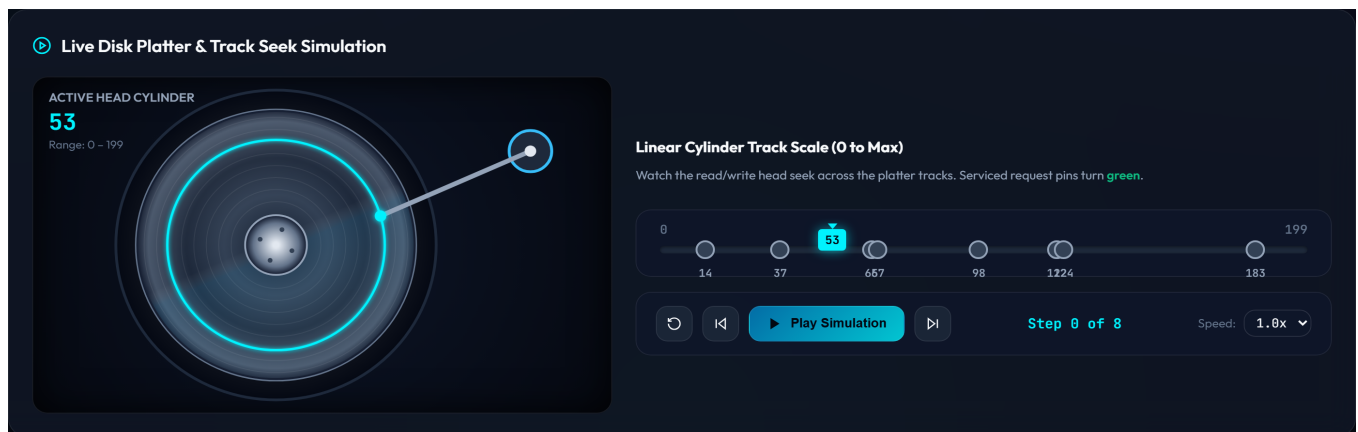
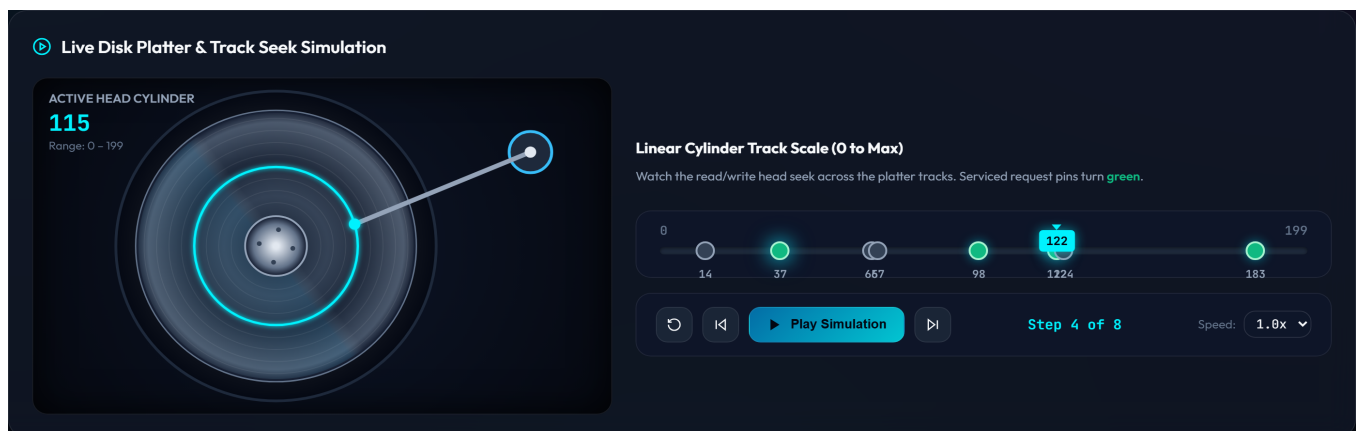


Figure 3: Live Playback in Progress (Active Head Seeking & Serviced Pins in Green)



Note: In Figure 3, the playback engine has completed 4 steps. The read/write head is actively positioned at track 37. Pins for serviced cylinders turn bright green, while pending requests remain amber.

6. Trajectory Plots & Comparative Visual Analytics

The classical Silberschatz trajectory graph plots **Disk Cylinder on the horizontal axis (0 to 199)** against **Servicing Order (Step 0 to N)** on the vertical axis. The slope and shape of the path illustrate mechanical efficiency.

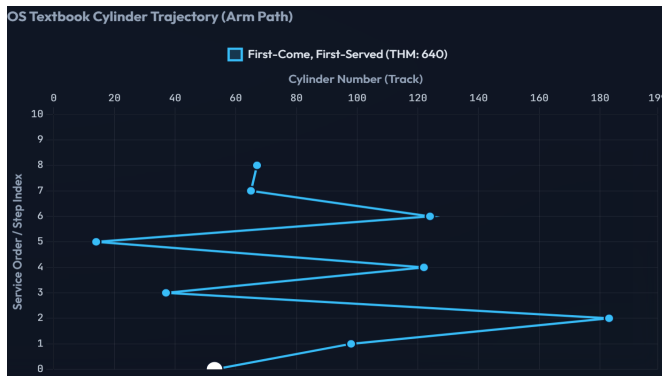


Figure 4: FCFS Trajectory (THM: 640)
High oscillation and erratic swings across platters.



Figure 5: SSTF Trajectory (THM: 236)
Localized greedy sweeps resulting in minimal head travel.



Figure 6: SCAN Trajectory (THM: 331)
Monotonic sweep to boundary 199 then reverse.

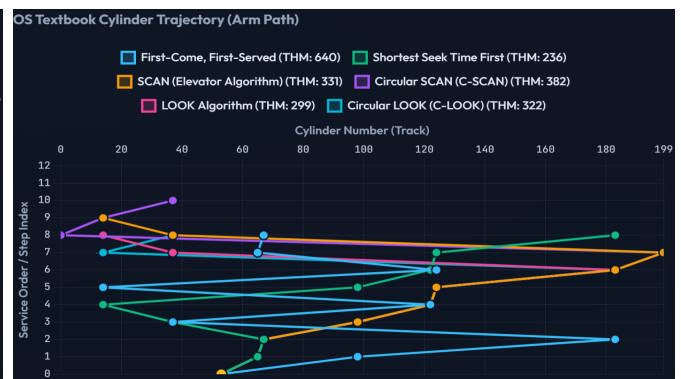


Figure 7: Multi-Algorithm Trajectory Overlay
Simultaneous comparison of all 6 arm pathways.

Figure 8: Comparative Metrics Bar Charts (Total Head Movement & Average Seek Length)



7. In-App Calculation Engine & Results Matrix Screenshots

Figure 9: In-App Live Mathematical Step-by-Step Calculation Table

Step-by-Step Head-Movement Calculations

Head Movement formula: $\text{Seek Distance} = |\text{Target Cylinder} - \text{Current Cylinder}|$. Active step during playback is highlighted in real time.

STEP #	FROM TRACK	TO TRACK	FORMULA	SEEK DISTANCE (ΔI)	CUMULATIVE MOVEMENT	OPERATIONAL RATIONALE
0 (Start)	—	53	Initial head position	0	0	Start Position
1	53	65	$ 65 - 53 $	12	12	Moving right/up towards cylinder 65
2	65	67	$ 67 - 65 $	2	14	Moving right/up towards cylinder 67
3	67	98	$ 98 - 67 $	31	45	Moving right/up towards cylinder 98
4	98	122	$ 122 - 98 $	24	69	Moving right/up towards cylinder 122
5	122	124	$ 124 - 122 $	2	71	Moving right/up towards cylinder 124
6	124	183	$ 183 - 124 $	59	130	Moving right/up towards cylinder 183
7	183	199	$ 199 - 183 $	16	146	Reaching disk boundary (track 199) and reversing direction
8	199	37	$ 37 - 199 $	162	308	Moving left/down towards cylinder 37
9	37	14	$ 14 - 37 $	23	331	Moving left/down towards cylinder 14

Figure 10: In-App Comparative Results Matrix & Workload Behavioral Analysis

Disk Scheduling Simulator & Analysis

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Comparative Results & Performance Analysis Matrix

Direct side-by-side benchmark comparing Total Head Movement, Average Seek Length, Starvation Vulnerability, and Algorithmic Complexity.

ALGORITHM	TOTAL HEAD MOVEMENT (THM)	AVERAGE SEEK LENGTH	EST. SEEK TIME (@5MS)	EFFICIENCY VS FCFS	SERVICE ORDER	STARVATION RISK	COMPLEXITY
First-Come, First-Served	640	80.00	3200.0 ms	Baseline	53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67	None	O(N)
Shortest Seek Time First ★ Most Efficient	236	29.50	1180.0 ms	+63.1% Faster	53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183	High (distant tracks can starve)	O(N²)
SCAN (Elevator Algorithm)	331	41.38	1655.0 ms	+48.3% Faster	53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 37 → 14	Low (uniform sweep)	O(N log N)
Circular SCAN (C-SCAN)	382	47.75	1910.0 ms	+40.3% Faster	53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 0 → 14 → 37	Very Low (uniform wait time)	O(N log N)
LOOK Algorithm	299	37.38	1495.0 ms	+53.3% Faster	53 → 65 → 67 → 98 → 122 → 124 → 183 → 37 → 14	Low	O(N log N)
Circular LOOK (C-LOOK)	322	40.25	1610.0 ms	+49.7% Faster	53 → 65 → 67 → 98 → 122 → 124 → 183 → 14 → 37	Very Low	O(N log N)

8. Comparative Analysis & Modern OS Engineering

Algorithm	Total Head Movement	Avg Seek Length	Est. Seek Time	Efficiency vs FCFS	Starvation Risk	Arm Wear Index
FCFS	640 cylinders	80.00 tracks/req	3,200.0 ms	Baseline (0.0%)	Zero (Fair)	Severe (High Stress)
SSTF	236 cylinders	29.50 tracks/req	1,180.0 ms	+63.1% Gain	High Risk	Moderate
SCAN	331 cylinders	41.38 tracks/req	1,655.0 ms	+48.3% Gain	Zero (Fair)	Low (Smooth Sweep)
C-SCAN	382 cylinders	47.75 tracks/req	1,910.0 ms	+40.3% Gain	Zero (Fair)	Low (Circular Sweep)
LOOK	299 cylinders	37.38 tracks/req	1,495.0 ms	+53.3% Gain	Zero (Fair)	Very Low (Optimal Range)
C-LOOK	322 cylinders	40.25 tracks/req	1,610.0 ms	+49.7% Gain	Zero (Fair)	Very Low (Optimal Circular)

Engineering Trade-Offs:

- Throughput vs Fairness:** SSTF provides maximal raw throughput (236 cylinders), but its starvation vulnerability makes it unsuitable for mission-critical enterprise systems without aging mechanisms. SCAN and LOOK provide bounded response times.
- Mechanical Stress:** FCFS creates abrupt reversals that accelerate mechanical wear on spindle bearings and voice coils. SCAN's monotonic sweeping motions significantly extend hardware lifespan.
- LOOK Superiority over SCAN:** LOOK eliminates SCAN's redundant travel to track 199, saving 32 tracks (nearly 10% gain).

Modern Operating System I/O Schedulers:

- Linux Deadline Scheduler:** Implements a SCAN/LOOK elevator sweep combined with FIFO expiration deadlines (e.g., 500 ms for reads, 5 s for writes) to strictly prevent starvation.
- CFQ & BFQ:** Allocate fair bandwidth budgets and time slices across competing processes.
- NVMe & Solid-State Storage (SSDs):** Because flash memory has no moving read/write heads, seek latency is zero. Modern SSD schedulers (*None*, *Kyber*) focus on multi-queue hardware submission queues rather than physical track reordering.

9. Conclusion

This project successfully designed, implemented, and empirically validated a full interactive disk scheduling simulation system. The quantitative benchmark confirmed that SSTF (236) and LOOK (299) achieve dramatic seek reductions over unoptimized FCFS (640), while SCAN variants deliver the most reliable balance between throughput, starvation avoidance, and hardware longevity.

10. Authentic Academic References

[1] Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating System Concepts* (10th Edition). Chapter 11: Mass-Storage Structure. John Wiley & Sons. ISBN: 978-1-119-32091-3.

[2] Tanenbaum, A. S., & Bos, H. (2015). *Modern Operating Systems* (4th Edition). Chapter 5: Input/Output. Pearson. ISBN: 978-0-13-359162-0.

[3] Stallings, W. (2018). *Operating Systems: Internals and Design Principles* (9th Edition). Chapter 11: I/O Management and Disk Scheduling. Pearson. ISBN: 978-0-13-467095-9.

[4] Bovet, D. P., & Cesati, M. (2005). *Understanding the Linux Kernel* (3rd Edition). Chapter 14: The Block Device Driver Architecture & I/O Schedulers. O'Reilly Media.

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